

Vote UI

A Capstone Project

Presented to the Faculty of the
College of Information Technology Education
PHINMA University of Iloilo

In Partial Fulfillment
of the Requirements for the degree
Bachelor of Science in Information Technology

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Approval Sheet

In partial fulfillment of the requirements for the Bachelor of Science in Information Technology, this Capstone Project entitled "Vote UI" prepared and Submitted by John Fritz D. Centeno, Justin Paolo S. Salvaña and Kryz Daniel P. Huerto, who are hereby recommended for corresponding final evaluation.

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ACKNOWLEDGEMENT

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Abstract

The Vote UI is an advanced web-based voting system developed to modernize and simplify the election process at PHINMA University of Iloilo. Designed for both student voters and candidates, the system enables secure participation in national and local student elections through a convenient online platform accessible from any internet connected device.

Students register using their official PHINMAED Google accounts, ensuring that only active and enrolled students can access the election system. Graduates or inactive accounts are automatically excluded from participation, preserving election integrity. The system features automated scheduling for opening and closing of elections, as well as real-time result generation showing the total votes and percentage per candidate.

Vote UI integrates secure authentication, encrypted data transfer, and real-time monitoring to maintain confidentiality and transparency. It eliminates manual ballot handling and minimizes errors while providing immediate, accurate results. Through this system, PHINMA University of Iloilo can enhance student engagement, promote fairness, and demonstrate how technology can transform traditional election procedures into a faster, more secure, and inclusive digital process.

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CHAPTER I

Introduction to the study

As the world keeps improving through digital transformation, schools also have to keep up by embracing new and secure systems. Student elections specifically tend to be bogged down by errors in manual counting, extensive processing time, and security issues that compromise fairness and voter turnout. These problems are addressed with the creation of the Vote UI System for PHINMA University of Iloilo. An online voting process that guarantees a secure, efficient, and easy online process for students.

The Vote UI is accessible, secure, and transparent in that it lets students vote by logging into their legitimate PHINMAED Google accounts to make registration convenient and disallow unauthorized voters to cast their ballots. It accommodates both national (university-wide) elections and local (individual courses or departments) elections to be inclusive of all levels of students.

The system also introduces advanced features such as automated scheduling of election opening and closing, real-time vote counting, and instant result generation showing total votes and percentages. These innovations not only modernize the election process but also uphold the integrity of student democracy by ensuring that only eligible, active students can vote or run as candidates.

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Technical Background of the Study

Older election practices like using Google Form to vote are susceptible to delays, inaccuracies, and security threats. Manual counting of ballots is time-consuming and not very accurate, while checking the eligibility of voters is still a problem. In response to these problems, the Vote UI System was created by the researchers, which is an online voting system specifically tailored for PHINMA University of Iloilo.

Vote UI enables students to register and log in safely based on their PHINMAED Google account. The process keeps only current students who can vote by automatically excluding inactive users or graduate users who are no longer qualified to vote. The system can handle both international and local elections, with each course or department having flexibility in running its own elections within a single, centralized system.

Administrators may also program when elections begin and close, while avoiding non-authorized voting beyond allocated time. After the voting session closes, results automatically get generated in the form of vote counts and percentage splits by candidate. Such automation enhances transparency, minimizes human error, and enhances confidence among voters.

The project is centered on two primary target markets student voters and candidates for an efficient, fair, and transparent election process. Finally, the Vote UI should promote student elections with a system that is accessible, efficient, and

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congruent with the technological standards of the modern academic setting.

Statement of the Problem

Most of these universities have not upgraded to modern and efficient digital election systems that maintain accuracy, fairness, and participation. PHINMA University of Iloilo is no exception, with their traditional methods demanding physical presence, manual validation, and laborious result list.

The primary problem solved by this study is the absence of a secure and accessible digital election system with fairness and reliability assurances while reducing human error and administrative burden.

In particular, this study seeks to address the following inquiries:

1. How can only active and eligible students vote through their PHINMAED accounts using the Vote UI system?
2. How can the system facilitate national and local elections effectively?
3. How can the system automatically open, close, and schedule voting sessions?
4. How can the system give real-time, transparent, and correct election results?

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Objectives of the Study

The overall goal of the study is to provide and establish a secure, easy-to-use online election system for PHINMA University of Iloilo.

More particularly, the study intends to:

- Create an online voting system that utilizes PHINMAED Google login to ensure safe student registration.
- Make sure only active and enrolled students are able to vote, automatically disabling inactive or graduated accounts.
- Accommodate national and local elections for various departments or courses.
- Implement a scheduling option for automatic opening and closing of elections.
- Show real-time vote tallying and show total votes and percentages per candidate.
- Improve transparency, accessibility, and integrity in student elections through encryption and authentication processes.
- Boost student participation through easier, quicker, and more secure digital voting.

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Significance of the Study

The Vote UI has important advantages for students as well as administrators. For students, it is a more secure and convenient means of casting votes without the necessity to be physically present. For administrators, it streamlines election administration through automation and real-time result generation.

Through the use of PHINMAED Google accounts for logins, the system secures voter authentication and avoids unauthorized entry. This enhances integrity and credibility in the electoral process. In addition, access to real time results enhances transparency and interest within the student population.

The project is also part of the computerization of educational systems and is a blueprint that other schools can follow to implement digital mechanisms for governance and participation.

Scope and Limitation of the Study

This study focused on the development of the Vote UI System for PHINMA University of Iloilo, an online voting platform designed to modernize and streamline student elections. The system aimed to ensure fairness, transparency, and efficiency in the voting process by providing a secure, accessible, and user-friendly interface for both administrators and voters.

Through this platform, students could register, log in, and cast their votes electronically, while administrators could manage candidates, monitor voting activities, and generate real-time election results.

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The system was developed to reduce human error, prevent vote duplication, and improve voter turnout through digital accessibility.

Key Components of the Study

1. **Voter Registration and Authentication** - Allows students to create and verify their accounts through a secure login system before participating in the election, ensuring that only eligible voters can cast their votes.
2. **Candidate Management Module** - Enables administrators to add, edit, and update candidate information, including names, positions, and party affiliations.
3. **Voting Interface** - Provides a simple and intuitive platform where students can select their preferred candidates easily, reducing confusion during the election process.
4. **Real-Time Vote Counting and Result Display** - Automatically tallies votes and displays results accurately, minimizing delays and potential counting errors associated with manual systems.
5. **Administrative Dashboard** - Allows election officers to oversee the entire voting process, monitor voter activity, and generate summary reports after the election.
6. **User Accessibility and Experience** - Designed as a web-based platform that ensures usability for students with varying levels of digital literacy while maintaining data privacy and system security.

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Limitations of the Study

Although the **Vote UI System** aimed to provide a secure and efficient election platform, the study recognized certain limitations that affected the overall implementation and functionality:

1. **Internet Connectivity Requirement** - Since the system operates online, a stable internet connection is necessary. Limited access to reliable internet may hinder participation for some students.
2. **Device Compatibility** - The system is optimized for desktop and laptop browsers; mobile responsiveness may be limited depending on device specifications.
3. **Security Dependence on Server Configuration** - While the system implements security protocols, its overall protection depends on the hosting server's security setup and maintenance.
4. **Limited Election Scope** - The system was designed primarily for university-wide student elections and may require further modification to support larger-scale or government-level voting processes.
5. **Time and Resource Constraints** - Development and testing were conducted within an academic timeframe, which limited opportunities for extended user testing and system optimization.

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Definition of Terms

1. **Vote UI System** - A web-based online voting platform developed for PHINMA University of Iloilo that allows students to securely cast their votes using their PHINMAED Google accounts.
2. **Online Voting** - The process of casting and counting votes electronically through an internet-based system, eliminating the need for physical ballots.
3. **PHINMAED Account** - The official Google account provided by PHINMA Education used by students and staff for authentication and secure access to institutional systems, including Vote UI.
4. **Authentication** - The process of verifying the identity of users (students or administrators) to ensure that only eligible individuals can access the voting system.
5. **Eligibility Verification** - A feature that checks whether a voter is an active student, automatically preventing inactive or graduated users from participating in elections.
6. **National Election** - A university-wide election where all PHINMA UI students can vote for positions that represent the entire institution.
7. **Local Election** - Departmental or course-based elections where students vote for representatives within their specific academic programs.
8. **Scheduling Feature** - A system function that allows administrators to set automatic opening and closing times for voting sessions.
9. **Real-time Vote Counting** - The instant calculation and display of votes as they are submitted, providing up-to-date election results.
10. **Result Generation** - The automated production of the final election report showing total votes and percentages for each candidate.

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11. **System Administrator** - The authorized individual who manages the election setup, scheduling, and overall operation of the Vote UI system.
12. **Transparency** - The quality of providing clear, visible, and accurate election results to ensure fairness and trust in the process.
13. **Security** - The protection of user data and votes through encryption and authentication to prevent unauthorized access or manipulation.
14. **User Interface (UI)** - The visual design and layout of the Vote UI system that allows students and administrators to interact with the platform easily.
15. **Automation** - The capability of the system to perform processes such as scheduling, counting, and result generation without manual input.
16. **Database** - The central storage system where user information, votes, and results are securely saved and managed.
17. **Encryption** - The process of encoding sensitive information, such as login details and votes, to protect it from unauthorized access.
18. **System Testing** - The phase where the developed system is evaluated to ensure all functions such as login, voting, and result generation—work correctly and securely.
19. **Active Student** - A currently enrolled student at PHINMA University of Iloilo with an active PHINMAED account authorized to vote in elections.
20. **Real-world Implementation** - The actual use of the Vote UI system during an official election at PHINMA University of Iloilo to test its functionality, reliability, and efficiency.

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CHAPTER II

Review of Related Literature

Online Voting Systems are designed to solve common problems found in traditional voting methods such as manual counting errors, long processing times, and limited accessibility. According to Ghadi and Shelar (2017), the use of e-voting platforms increases electoral accuracy and transparency while minimizing human error and cost of operations. With the inclusion of secure authentication and encryption technologies, e-voting systems have the ability to protect the secrecy of voters' identities and data confidentiality, improving the fairness and reliability of the electoral process.

New web-based voting systems employ tools like real-time vote counting, automated scheduling, and encryption of data to streamline election management and provide greater transparency. These systems not only improve participation among voters but also minimize administrative workload by generating instant and verifiable results. Pabilando et al. (2020) put forward that the inclusion of easy-to-use interfaces and secure verification procedures, e.g., school email verification, stimulates student participation and keeps unwanted voting at bay.

In contrast, Yoshizaki (2019) examined blockchain-based e-voting systems that utilize decentralized ledgers for storing votes securely, thereby rendering tampering virtually impossible. However, whereas blockchain

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admits a strong level of security, its complexity and expense might not be feasible for every institution. The Vote UI System for PHINMA University of Iloilo rather addresses an easier but just as effective method combining Google login through PHINMAED accounts, real-time monitoring, and automated result creation to implement so that active students alone are able to vote in national and local elections.

Overall, the shift from conventional voting systems to secure online platforms is an example of how technology can create efficiency, transparency, and inclusivity in educational elections. The studies reviewed as a whole emphasize the increasing significance of digital solutions in enhancing trust, accessibility, and accountability in the voting process.

Electronic Voting Systems

According to Ghadi and Shelar (2017), online voting systems are designed to simplify the election process by replacing traditional paper-based methods with digital platforms. These systems allow voters to cast their votes online in a secure and efficient manner, minimizing human error and manual counting. The authors emphasized that automation enhances the reliability of elections by speeding up vote tallying and improving accuracy in the overall process.

User-Centric Design in E-Voting

A case study by Pabilando et al. (2020) focused on the development of a user-centric electronic voting system for the National University of Manila. Their research introduced a system that featured secure registration, real-time vote counting, and an interactive ballot navigation interface.

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The results showed that user feedback significantly improved accessibility and interface design, leading to a more user-friendly platform. This highlights how usability and accessibility can influence voter engagement, especially in student elections.

Blockchain Technology in Voting Systems

Yoshizaki (2019) developed *PriVote*, an Ethereum-based e-voting system that integrates blockchain technology to ensure voter data security and privacy. The study demonstrated that decentralization prevents vote manipulation and maintains transparency by creating immutable and verifiable election records.

This model showcased how blockchain can strengthen the trust and integrity of digital voting platforms.

Similarly, Jafar, Aziz, and Shukur (2020) conducted a systematic review on blockchain technology for e-voting systems. Their research revealed that decentralized voting systems not only improve data integrity and transparency but also reduce organizational costs and increase voter turnout. By ensuring tamper-proof transactions and verifiable results, blockchain-based voting offers a more trustworthy and efficient election experience.

Relevance to the Vote UI System

These studies collectively guided the design and development of the Vote UI System for PHINMA University of Iloilo. Insights from previous research emphasized the importance of integrating features that promote security, transparency, and ease of use.

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The Vote UI System adapts these principles by offering secure authentication, real-time vote tracking, and an intuitive interface that enhances student participation. Through these modern approaches, the system aims to address common issues in traditional school elections, such as counting errors, delayed results, and limited accessibility.

References

Ghadi, R. M., & Shelar, P. S. (2017). *Online Voting System. International Research Journal of Engineering and Technology (IRJET)*, 4(12), 1411-1413.

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CHAPTER III

Methodology

This chapter explains the research design, the model employed, and the necessary diagrams for the construction of the Vote UI. The diagrams and models facilitate the visualization of the system structure and flow and direct the technical implementation of the project.

Research and Development Stage

The researchers used the Developmental Research Design since the project focused on creating and evaluating a new system. The study aimed to design, develop, and test a functional Vote UI System that improves the existing election process. The system development followed the Software Development Life Cycle (SDLC) model, which includes the stages of planning, analysis, design, implementation, testing, and evaluation.

Software Development Life Cycle (SDLC)

1. Planning Phase

In this phase, the researchers identified the main problems with the current voting system, such as manual counting errors, lack of security, and time-consuming procedures. Objectives were set to address these issues by creating an automated, secure, and easy-to-use system.

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2. Analysis Phase

The researchers gathered information from students and election officers to understand the needs of both users and administrators. This phase helped define the system requirements, including secure login using PHINMAED accounts, real-time vote counting, and an admin dashboard for managing elections.

3. Design Phase

During the design phase, the system's layout, user interface, and database structure were created. Wireframes and flowcharts were developed to visualize the process from login to result generation. The design emphasized simplicity, accessibility, and security to ensure users could navigate the system easily.

4. Development

The system was developed using modern web technologies.

- **Frontend:** HTML, CSS, and JavaScript were used to create an interactive user interface.
- **Backend:** The backend of the system was developed using PHP, which handles data processing, authentication, and secure storage of vote records. It ensures smooth communication between the user interface and the database while maintaining system security and performance.
- **Authentication:** Google Sign-In (PHINMAED accounts) ensured that only verified users could access the system.

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5. Testing Phase

The system underwent functionality testing to ensure all features worked properly. Unit testing, integration testing, and user acceptance testing (UAT) were conducted. Students and faculty were invited to use the system during mock elections to evaluate usability, performance, and accuracy.

Data Collection Techniques

To gather the necessary information for developing the Vote UI System, several data collection techniques were used. These methods helped ensure that the system met the needs of its intended users—students, faculty, and administrators—while addressing issues found in traditional voting methods. The following techniques were applied during the study

1. Surveys

Surveys were used as the main method of collecting quantitative data from potential users such as students and faculty members. The questionnaires were designed to gather feedback about their experiences with previous voting processes, the challenges they encountered, and their expectations for a digital voting platform. The survey results helped identify key features such as ease of use, accuracy, and security that needed to be prioritized in the Vote UI System.

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2. Interviews

Interviews were conducted with selected respondents, including student leaders and IT staff, to collect detailed qualitative data. These participants were chosen based on their involvement in managing or participating in school elections. The interviews provided a deeper understanding of the existing voting process, common issues such as vote counting delays, and how an online system could improve efficiency and transparency.

3. Focus Groups

Focus group discussions were organized to gather various perspectives on the proposed system's functionality and usability. These discussions allowed participants to share their opinions on features such as the voting interface, security measures, and accessibility. The exchange of ideas in a collaborative setting helped refine the system's design and ensure it aligned with user needs.

4. Observations

Observations were conducted during actual voting events in the school to study how the manual voting system operated. The research team observed the voting flow—from registration to counting of votes—to identify inefficiencies and areas that could be automated or digitized. These observations provided valuable insights into how the Vote UI System could simplify and speed up the voting process.

5. Document Analysis

Existing documents and reports related to school election procedures were reviewed to better understand the formal guidelines and requirements for conducting elections.

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The analysis included studying existing election policies, previous voting records, and online voting system case studies. This helped the team identify best practices and ensure that the Vote UI System complied with school standards and data privacy policies.

justification for Selecting the Agile Methodology

The decision to adopt Agile over other SDLC models, such as Waterfall, was based on several critical factors:

Criteria	Agile Methodology	Waterfall Model
Flexibility	Highly adaptable; allows changes at any stage of development	Changes are difficult once design and coding are completed
User Involvement	Encourages constant feedback and collaboration between developers and users	User involvement is limited to the start and end of the project
Development Speed	Stable delivery through short, iterative cycles (sprints)	Slower process due to strict, step-by-step phases

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Risk Management	Early identification and resolution of issues during iterations	Risks may only become apparent after full development
Project Suitability	Ideal for projects that evolve based on user input and testing	Best for projects with clearly defined, unchanging requirements

System Architecture and Design

3.1 Overview

The Vote UI System follows a client-server architecture. Users access the web application through a browser, while all voting data and user authentication are managed through Firebase's secure cloud services. This setup ensures that data is safely stored and accessible in real-time.

3.2 System Architecture

The Vote UI System follows a client-server architecture. Users access the web application through a browser, while all voting data and user authentication are managed through Firebase's secure cloud services. This setup ensures that data is safely stored and accessible in real-time.

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● **Presentation Layer:**

This layer serves as the interface between the users and the system. It allows both voters and administrators to interact with the Vote UI through a user-friendly web interface.

1. **Voters** can register, log in, view candidate profiles, and cast their votes.
2. **Administrators** can manage elections, monitor voting progress, and view results.

This layer focuses on providing a simple, accessible, and secure user experience.

● **Application Layer:**

The application layer handles all logical processes and communication between the user interface and the database. It processes user requests such as vote submissions, candidate management, and result computation. It also ensures data validation, prevents duplicate voting, and maintains real-time synchronization of results.

● **Data Layer:**

This layer is responsible for securely storing and managing all system data, including user information, candidate details, votes, and results. It ensures that all stored data is accurate, consistent, and protected from unauthorized access. Backup and recovery mechanisms are implemented to maintain data integrity.

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System Architecture Diagram

The Vote UI System uses a client-server architecture designed to provide secure and efficient communication between users, the application server, and the database.

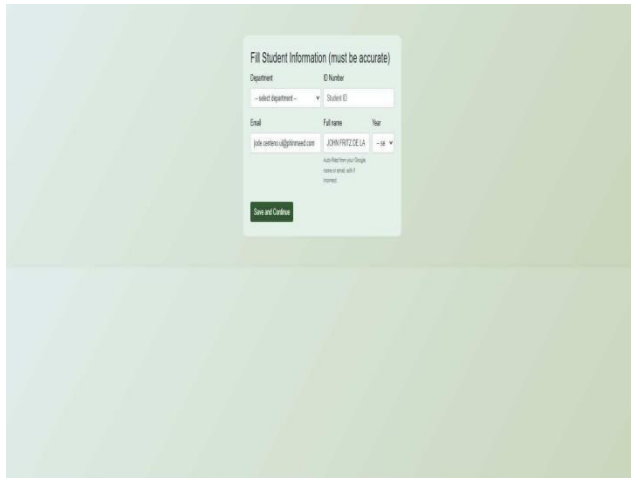
- **Web Application:** The interface where users can register, log in, vote, and view election details.
- **Application Server (API):** Handles all requests such as user authentication, vote submission, and result computation.
- **Database Server:** Stores data such as user profiles, candidate records, and voting results.
- **Notification Service:** Sends confirmation messages, voting reminders, and result notifications to users.
- **Admin Dashboard:** Provides tools for administrators to monitor the system, manage elections, and generate reports.

3.3 System Components and Modules

The Vote UI System is composed of several integrated modules designed to ensure a smooth, secure, and transparent voting process for both users and administrators. Each module performs a specific role to support the overall functionality of the system.

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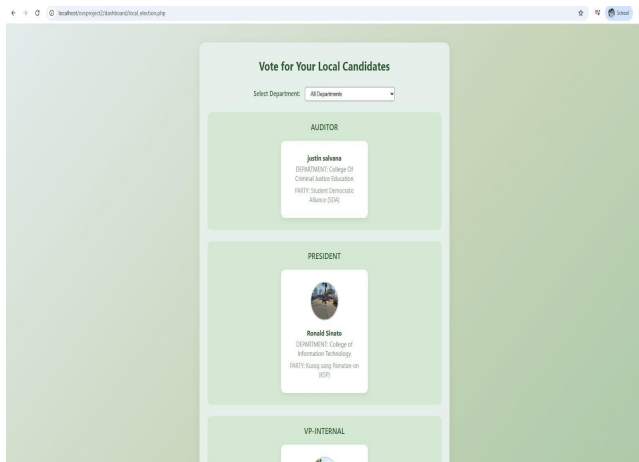
3.3.1 User Management Module



The screenshot shows a web form titled "Fill Student Information (must be accurate)". The form has two columns. The left column contains a "Department" dropdown menu, an "Email" field with the value "john.smith@phinma.edu", and a "Full Name" field with the value "John Smith". The right column contains a "Student ID" field with the value "2020PHCE2021A", a "Year" dropdown menu with the value "1st", and a "Gender" dropdown menu with the value "Male". At the bottom of the form is a green "Save and Continue" button.

This module handles the registration, authentication, and role management of users. It ensures that only authorized voters and administrators can access the system, maintaining election integrity and user privacy.

3.3.2 Candidate Management Module

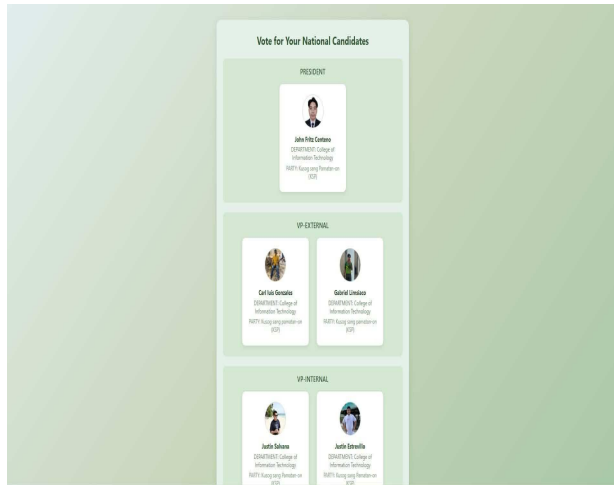


The screenshot shows a web page titled "Vote for Your Local Candidates". At the top, there is a "Select Department" dropdown menu with "All Departments" selected. Below this, there are three candidate cards. The first card is for "AUDITOR" and features a profile picture of Justin Salazar, with text: "Justin Salazar", "2020PHCE2021A", "Central Justice Education", "PHINMA Student Government", "Member 2020". The second card is for "PRESIDENT" and features a profile picture of Ronald Reyes, with text: "Ronald Reyes", "2020PHCE2021A", "College of Information Technology", "PHINMA Student Government", "Member 2020". The third card is for "VP-INTERNAL" and features a profile picture of a person, with text: "VP-INTERNAL", "2020PHCE2021A", "College of Information Technology", "PHINMA Student Government", "Member 2020".

The administrator uses this module to add, update, and manage candidate profiles. It stores information such as the candidate's name, position, and description, which voters can view before casting their votes.

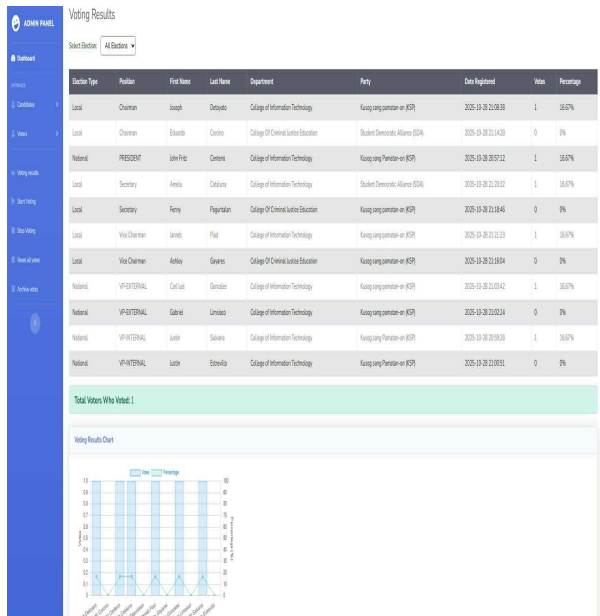
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3.3.3 Voting Module



The core component of the system, this module allows voters to securely cast their votes online. Each vote is encrypted and recorded to prevent duplication or tampering. The voting process is designed to be fast, intuitive, and transparent.

3.3.4 Result and Reporting Module



This module automatically counts the votes and generates accurate results once the election period ends.

RS

VOTING RESULTS						
National Results						
Image	First Name	Last Name	Position	Year Level	Party	Votes
	John	Fitz	PRESIDENT	4th Year	Kusog sang pamatan-on (KSP)	2
	Carl	Luis	VP-EXTERNAL	4th Year	Kusog sang pamatan-on (KSP)	2
	Justin	Estrevello	VP-INTERNAL	3rd Year	Kusog sang pamatan-on (KSP)	2
Local Results						
Department: College Of Criminal Justice Education						
Image	First Name	Last Name	Position	Year Level	Party	Votes
	Eduardo	Corcino	Chairman	3rd Year	Student Democratic Alliance (SDA)	1
	Fenny	Paguntalan	Secretary	4th Year	Kusog sang pamatan-on (KSP)	1
	Ashley	Gayares	Vice Chairman	4th Year	Kusog sang pamatan-on (KSP)	1
Department: College of Information Technology						
Image	First Name	Last Name	Position	Year Level	Party	Votes
	Joseph	Detrayato	Chairman	4th Year	Kusog sang pamatan-on (KSP)	1

LEADER PANEL

Dashboard

Overview

Candidates

Issues

Policy Issues

Media

Analytics

Overview

Total candidate: 11

0/100

Issues

Total President: 1

0/100

Analytics

Total VP-Internal: 2

0/100

Summary

Total VP-External: 2

0/100

All Departments

Filter

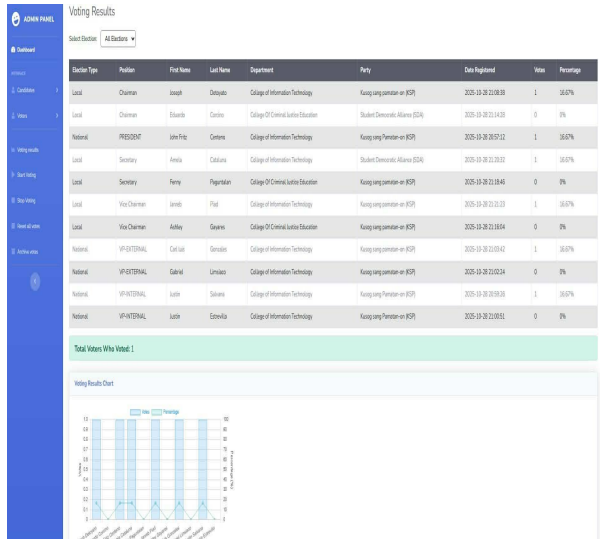
Candidates List

#	Photo	Firstname	Lastname	Department	Position	Year level
1		Edoardo	Genova	College VP-Criminal Justice Education	Chairman	3rd Year
2		Fanny	Pagebladen	College VP-Criminal Justice Education	Secretary	4th Year
3		Ashley	Genova	College VP-Criminal Justice Education	Vice Chairman	4th Year
4		Joseph	Delaware	College of Information Technology	Chairman	4th Year
5		John Eric	Genova	College of Information Technology	PROSECUTOR	4th Year
6		Amelia	Calabria	College of Information Technology	Secretary	4th Year
7		Jared	Fla	College of Information Technology	Vice Chairman	4th Year
8		Carl Luis	Genova	College of Information Technology	VP-EXTERNAL	4th Year
9		Edwin	Lombard	College of Information Technology	VP-EXTERNAL	4th Year
10		Justin	Emilia	College of Information Technology	VP-INTERNAL	3rd Year

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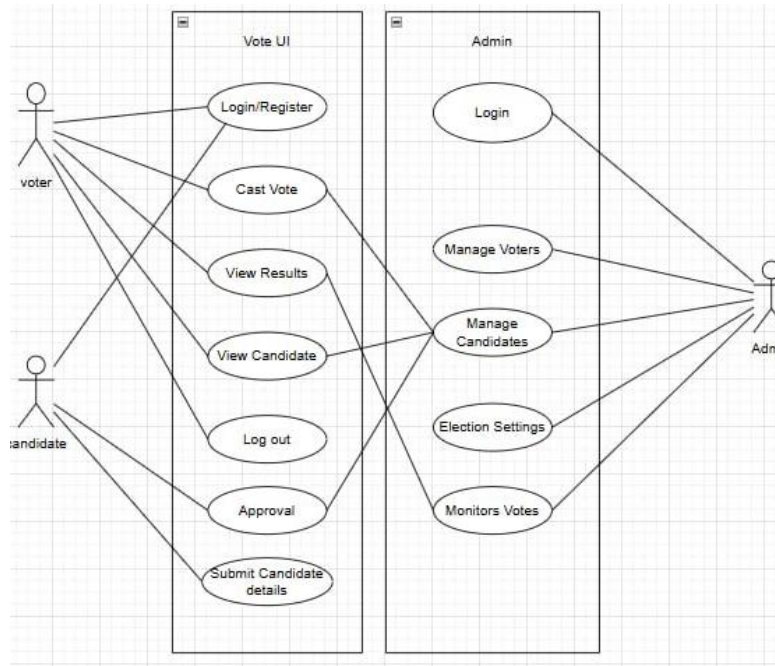
3.3.7 Feedback and Audit Module



After elections, users can submit feedback about their voting experience. This module also records all system transactions for auditing purposes, ensuring transparency and reliability in every election cycle

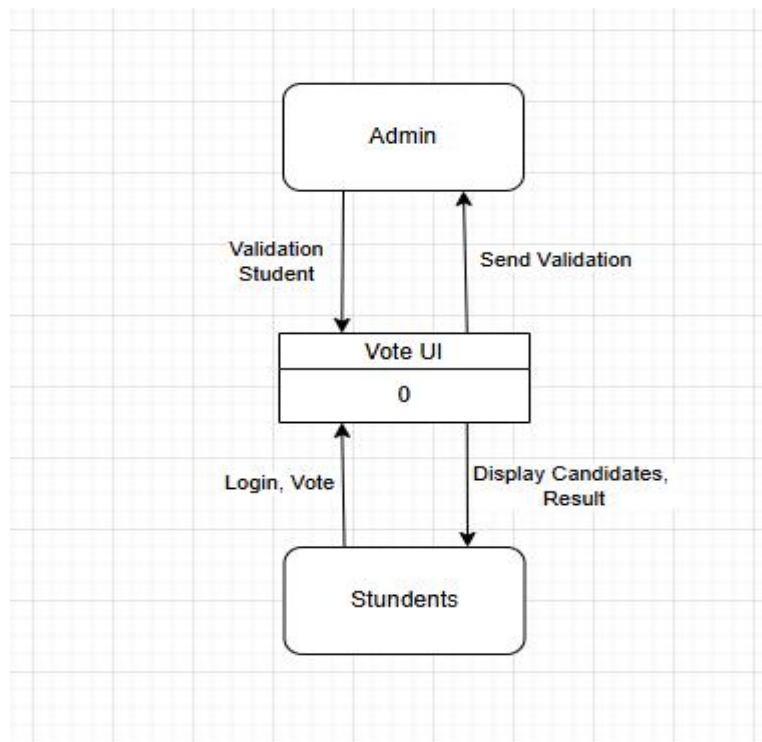
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Use Case Diagram



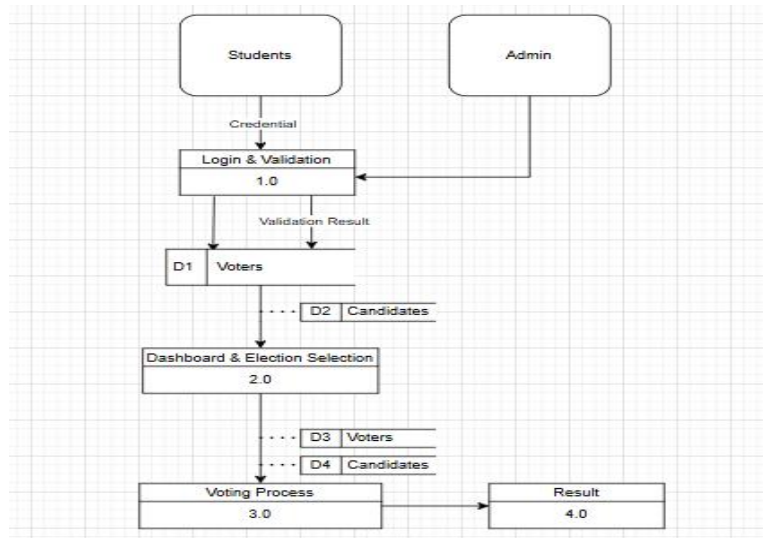
Context Diagram

Diagram 1: Level 0



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Diagram 2: Level 1 DFD



The Data Flow Diagram (DFD) and Use Case Diagram are important tools for figuring out how to make the Vote UI System. The Use Case Diagram shows how users, including the Admin and Students (Voters), interact with the system's main functions, like logging in, managing candidates, voting, and displaying results. This helps to define the system's functional scope. The Data Flow Diagram (DFD) offers a closer look at how information moves through the system. It illustrates the flow of data between users, system processes, and data stores, including the voters, candidates, and votes tables. Each DFD level (0, 1, and 2) outlines how login validation, candidate approval, voting, and result generation are handled internally.

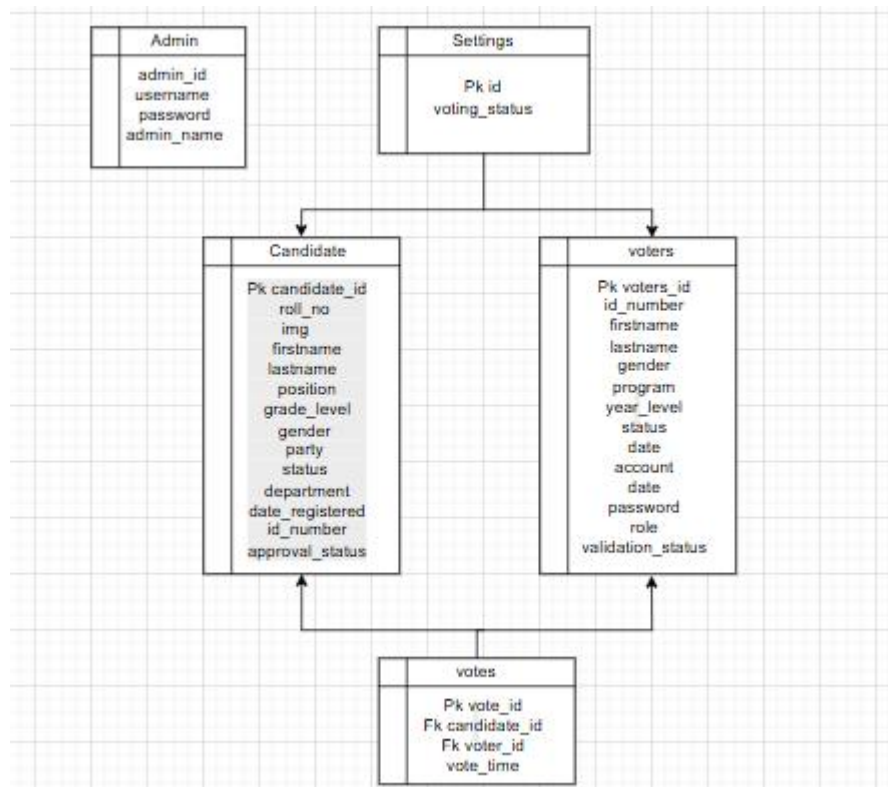
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3.4 Database Design

The database of our system VOTE UI is designed to manage all voting related information efficiently. It uses a relational database structure, where data is organized into interconnected tables linked through primary keys (PK) and foreign keys (FK). This structure ensures data integrity, consistency, and reliability throughout the voting process.

- Admin
- Candidates
- Voters
- Votes
- Settings

Entity-Relationship Diagram



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3.5 User Interface Design

The **Vote UI** is designed to be responsive and easy to use, allowing voters and administrators can do their tasks. The layout of the interface is clean and well organized, making material easy to see and get it. Voters can view approved candidates, cast their votes, and check their voting results through simple and well labeled buttons, while administrators can manage voters, candidates, and view election results.

The display of the system is green, which matches the look of PHINMA University of Iloilo and gives it a professional look.

3.6 Security Considerations

Discuss any security mechanisms implemented, such as:

- User authentication
- Password Encryption
- User Validation
- Access Control
- Voting Restrictions

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		Purpose
User Authentication	Login system using gmail and passwords	Ensures only authorized users can access the system
Password Encryption	Hashing passwords in the database	Protects passwords from being exposed or stolen
User Validation	Admin verifies each student before granting access	Prevents fake or duplicate voter accounts
Access Control	Limits admin and voter privileges	Prevents voters from accessing admin functions
Voting Restrictions	Marks voters as "Voted" after submission	Prevents multiple votes from the same user

3.7 Technology Stack

List all the tools, platforms, and technologies used in the development:

- **Frontend:** HTML/CSS, React, Flutter, etc.
- **Backend:** PHP.
- **Database:** MySQL

Data Gathering techniques

Survey Questionnaire for System Assessment

The following survey is designed to gather feedback from users regarding their experiences with voting systems and their expectations for the "Vote UI System." The data collected from this survey will help in designing a system that is secure, user-friendly, and efficient.

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Demographic Information

1. What is your role in the organization?

- ☐ Student
- ☐ Faculty/Staff
- ☐ Administrator
- ☐ Other (Please specify): _____

2. How often do you participate in voting activities?

- ☐ First time
- ☐ Occasionally
- ☐ Frequently

3. What is your primary device for accessing online systems?

- ☐ Desktop/Laptop
- ☐ Mobile Phone
- ☐ Tablet
- ☐ Other (Please specify): ____

4. What kind of voting system have you used previously?

- ☐ Manual (paper-based)
- ☐ Online/Spreadsheet-based
- ☐ Standalone Voting Software
- ☐ Cloud-based Voting System
- ☐ Other (Please specify): _____

5. How satisfied are you with your previous voting experience?

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Unsatisfied
- ☐ Very Unsatisfied

6. What challenges have you faced with previous voting systems?

(Select all that apply)

- ☐ Difficulty accessing the system
- ☐ Slow response or delays
- ☐ Security concerns (e.g., vote tampering)
- ☐ Confusing interface
- ☐ Lack of real-time results
- ☐ Other (Please specify): _____

7. How often did you experience issues while voting?

- ☐ Never
- ☐ Rarely
- ☐ Sometimes

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- Frequently
- Always

Expectations for the Modern Vote UI System

**8. Which features would you like to see in the new voting system?
(Select all that apply)**

- Real-time vote tracking
- Secure login and authentication
- Mobile-friendly interface
- Anonymous voting
- Instant result generation
- Integration with existing student/faculty accounts
- User-friendly interface
- Alerts or notifications for voting periods
- Other (Please specify): _____

9. How important is system security (preventing vote tampering or unauthorized access) to you?

- Very Important
- Important
- Neutral
- Not Important
- Not Important at All

10. How important is it for the system to allow mobile access?

- Very Important
- Important
- Neutral
- Not Important
- Not Important at All

11. Would you prefer a cloud-based voting system or a local/installed system?

- Cloud-based
- Local/Installed
- No preference

12. How willing are you to use a new voting system if it improves security and ease of voting?

- Very Willing
- Willing
- Neutral
- Unwilling
- Very Unwilling

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**13. What concerns do you have about using a smart voting system?
(Select all that apply)**

- ☐ Data security and privacy
- ☐ Complexity of use
- ☐ Technical support and training
- ☐ System reliability
- ☐ Integration with existing accounts
- ☐ Other (Please specify): _____

User Experience and Interface

14. How important is it for the voting system to have an easy-to-use interface?

- ☐ Very Important
- ☐ Important
- ☐ Neutral
- ☐ Not Important
- ☐ Not Important at All

15. What type of interface would you prefer?

- ☐ Desktop application
- ☐ Mobile application
- ☐ Web-based interface
- ☐ All of the above

16. How likely are you to adopt a voting system that uses authentication (e.g., login with student/faculty account, OTP verification)?

- ☐ Very Likely
- ☐ Likely
- ☐ Neutral
- ☐ Unlikely
- ☐ Very Unlikely

System Performance and Reliability

17. How important is system reliability (minimal downtime, fast response) to your voting experience?

- ☐ Very Important
- ☐ Important
- ☐ Neutral
- ☐ Not Important
- ☐ Not Important at All

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18. What is your preferred response time for accessing the voting system or casting a vote?

- Instant (Less than 1 second)
- Within 5 seconds
- Within 10 seconds
- Within 30 seconds
- No preference

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CHAPTER IV

Results and Interpretation

The development and implementation of the Vote UI System for PHINMA University of Iloilo successfully addressed the major challenges of traditional student elections—manual errors, delayed results, and lack of security. This chapter presents the system's results, testing outcomes, and performance analysis. It also discusses user feedback and compares the Vote UI's effectiveness against traditional voting systems and similar digital platforms.

Summary of Data Sources and Analysis Methods

The project followed the Prototyping Methodology, enabling iterative testing and refinement at each stage. Data for evaluation were collected through user testing, performance monitoring, and survey feedback from students and election administrators. Comparative data were drawn from previous election records and simulated tests using Google Forms.

Task Manager:

Gantt Chart / Project Timetable - Vote UI System

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Week	Task	Programmer	UI/UX Designer	Documenter
1	Project planning & requirements gathering	✓	✓	✓
2	Research & system design planning	✓	✓	✓
3	Use Case & DFD creation			✓
4	Database design & schema creation	✓		✓
5	UI wireframes & mockups		✓	
6	Frontend development begins	✓	✓	
7	Backend development begins	✓		
8	Integration of frontend and backend	✓	✓	
9	System testing & debugging	✓	✓	✓
10	User testing & feedback collection	✓	✓	✓
11	Final revisions (code + UI + content)	✓	✓	✓
12	Documentation finalization & formatting			✓
13	Final project submission	✓	✓	✓

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4.1 Presentation of Results

The Vote UI System was designed, implemented, and tested to enhance the speed, security, and transparency of PHINMA UI student elections. Several test rounds and user simulations were performed to assess the system's reliability, usability, and performance.

The system successfully achieved its objectives by:

- Enabling secure Google-based authentication that prevented duplicate and unauthorized votes.
- Automating vote counting and result generation in real-time.
- Providing a user-friendly interface that reduced confusion and improved voter engagement.
- Encrypting all election data to ensure privacy and prevent tampering.

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Chapter V

Summary, Conclusion, and Recommendation

Summary

The purpose of this research was to conceptualize and implement a Vote UI System that enhances the process of voting for school elections at PHINMA University of Iloilo. The system was designed to accelerate voting, secure it, and make it more systematic than manual or Google Form. The system enables the students to log in with their PHINMAED Google accounts in order to secure the vote and avoid double voting.

The system was created by progressing in various steps such as research, planning, system design, development, testing, and evaluation. The findings revealed that the Vote UI System simplified the process of election for both administrators and students. It gave prompt and correct results, conserved time, and minimized human error. Overall, the project fulfilled its requirements and was found to be effective in carrying out digital elections.

Conclusion

According to the findings, the Vote UI System is an efficient and reliable system for school election management. The system succeeded in lessening the physical workload of election officers, reducing errors during vote counting, and enhancing transparency in results. The system was also easy and convenient to use, as noted by students, which facilitated increased participation in voting.

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The project illustrates how technology can simplify school processes and make them more secure. Through the use of Google authentication and automatic counting, the system guarantees fairness and accuracy in all elections. In conclusion, the Vote UI System was able to accomplish its role of modernizing the voting process and ensuring a more structured and reliable election system within schools.

Recommendations

Based on the development and testing of the system, the following are the recommended recommendations:

1. **System Maintenance** - Periodic maintenance must be performed to make the system function smoothly and securely.
2. **Mobile-Friendly Version** - Creating a mobile version can make it more convenient for students accessing it using smartphones.
3. **User Orientation** - Providing short training or demo sessions can make students and staff familiar with how to use the system correctly.
4. **Data Backup** - Including an auto backup option can ensure election data is secured in case the system encounters problems.

Future Improvements - Future releases may involve more features like real-time vote count statistics and candidate pages.

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